**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 224 · Trees and binary search trees: implementation

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does a null Next mean in a singly linked list?

Q2. In a BST, where does a smaller key go?

Q3. What is the base case of a recursive insertion reaching an empty child?

Q4. Insert 10,20,30 in that order into an empty unbalanced BST. What shape results?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

binary search tree — Binary tree in which lower keys occur to the left and higher to the right.

root — Topmost tree node.

leaf — A node with no children.

in-order traversal — Visit left subtree, node, then right subtree.

successor — Next key in in-order sorted sequence.

## New learning

Find the starting node — The root is 40, so compare 25 with 40 and move left.

Repeat comparisons — 25>20 goes right to 30; 25<30 goes left.

Insert at a null child — Create node 25 at the empty left-child position of 30. This implementation ignores duplicate keys: when key equals node.Key, neither subtree changes.

## Worked example

Insert 12 into the BST 20 (root), 10 (left), 30 (right).

1. Predict first — Compare 12 with 20, then 10.

2. Model the reasoning — Insert 12 as the right child of 10, preserving the BST property.

# We Do · Practise with guidance

Insert 35 into BST root 25, left 15 and right 40.

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35 > 25 → right
35 < 40 → left
```

Predict first

35>25 leads right from the root.

Model the reasoning

35<40 leads left, placing 35 as the left child of 40.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Insert 12 into BST root 15, left 9 and right 20. Identify its parent and position.

### Question 2

Insert 50,30,70,20,40,35 into an empty BST. State parent and side of 35, and the in-order visit sequence.

### Question 3

In BST 40 with children 20 and 60, and 20 with children 10 and 30, delete leaf 10. Explain a second case in which deletion is more involved.

# Exit Ticket · Answer before checking

In-order traversal of a BST with distinct keys visits values in ascending order.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. It marks the end of the list.

Q2. Into the left subtree.

Q3. Create and return a new node at that null position.

Q4. A right-leaning chain 10→20→30.

### You Do pages · Independent questions

Q1. [4] 12<15 [1], moves left to 9 [1], 12>9 [1], so it is right child of 9 [1].

Q2. [6] Parent of 35 is 40 [1], it is left child [1]; in-order sequence 20,30,35,40,50,70 [4].

Q3. [6] Find node 10 by moving left twice [2]; it is a leaf [1], so set parent 20.Left=null [1]. For a node with two children, replace it using an in-order successor/predecessor and remove the replacement from its old location [2].

### Exit Ticket · Model answer

True. Left subtree, node and right subtree produce increasing keys.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**